Sri Lanka Institute of Information Technology



IT3060 - Human Computer Interaction

Assignment 1

User Research Techniques

IT3060HCI2022_Assinment1_Group01

COLOSSAL

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Group Member Details

Name	Student ID	Workload Distribution
B.D.R.A.M.A.K.B. Ranasinghe	IT20649494	• Magazine Page • Return Policy • Account (CRUD)
M.W.K.G.B. Thennakoon	IT20666842	• Sinhala books • About us • Address (CRUD)
D.G.K.P. Kumara	IT20637682	• Second hand books • E-Gift card • Checkout (CRUD)
R.D.D. Sampath	IT20646592	• Order • FAQ • Home page (CRUD)
W.M.S.L.K. Welgama	IT20648640	• Discount books • Contact us • Cart (CRUD)

Project Description

The software chosen to redesign by our team is Books.lk website. That is an E-commerce system. Books.lk is a one of largest web site which selling variety of books, magazine and CD/DVD such as Sinhala books, secondhand books, magazine, and E-Gift cards for children as well as adults and teenagers.

The purpose of this website is to facilitate online product selecting, ordering, payment and selling.

Usability may be the most vital indicator of websites' effectiveness as it plays a crucial role in the interaction with customers. Official Books.lk website has several usability issues. So, our aim is to redesign the website by avoiding those usability issues and improve the user experience.

- No attractive color themes.
- Messy navigation bar.
- No alignment.
- More whitespaces.
- Header is not well organized.
- No footer.

Alternative Considered

Website	Reasons why we selected
https://torontocupcake.com/index.html	• Repeated logos and symbol of the business is not clear. • The theme is not clear and there is not a proper color or font combination. • Not user friendly. • Text are not clearly visible. • Lot of unnecessary white spaces all over.
https://www.art.yale.edu/	• No alignment. • Not use proper color theme. • Hard to understand the content. • No proper header footer. • Not well-organized content.
https://www.plerdy.com/blog/20-sajtov-s-uzhasnym-juzabiliti/	• Hard to understand the design of the pages. • Content is not organized • The colors, textures, animations, and fonts are overly dissonant • Same font size uses all over. • Not very attractive color combination.
http://www.exmouth-view.co.uk/	• Pages are not properly design. • Can't find the navigation bar. • There are different font sizes and font colors in the pages. • Not categorized well. • There are no images for some contents.
https://www.riversideartcenter.org/home	• Site name does not inspire art lovers. • Logo style is outdated and does not attract users. • Background and theme are a little dull. • Used same font in all the pages. • The layout is not as cohesive as it should be • There is no comment section under each art, so artists can't get reviews.

User Groups Identify

1. Primary Users = those are the direct access users of the system. The user has frequent hand on experiences of the system.

   Books.lk primary users = Unregistered customer, buyer, Admin, Product Manager

2. Secondary Users = This user is using the system occasionally, it's like a those who use the system using another person's support.

   Books.lk secondary users = Business analysis, Researcher, Fashion designer

3. Tertiary Users = Those who affected by the introduction of the system. This user will influence its purchase.

   Books.lk tertiary users = Reseller, Intermediary sellers, Student, Other book shop sellers

User Personas

User 1- Prabhani Bhagya

PRABHANI BHAGYA

Chemical Analyst

About Me

I am a chemical analyst over 2 years in SGS Lanka and graduated from Horizon campus in 2019.
I love read books since my childhood.
I am eager to learn new things daily throughout the world. By reading books, one avoids being limited to only one limit. As a hobby I read new and old books. Through that I can get a great understanding not only about my subject but also about other visions.

Work Experience

- 2019 - 2022 : Chemical Analyst
  SGS Lanka pvt.Ltd

Goals

- Looking for international books that related to Bio technological and chemistry .
- Enhancing knowledge of education.
- Discover newly released books and translations.
- Order required books and spend my free time with books.

Strengths

Flexibility, Dedication, Collaboration, Patience

Interest

International Chemistry Books
Translated educational books all over the world.
Magazines
Novels
English Books



Female

Age - 26 Years

prabhu@gmail.com

3A, Annie avenue,
Dehiwala,
Colombo,
Sri Lanka.

Pain Points

- Newly released biological books, chemistry books are hard to find on this site.
- There is no images for some related categories.
- Content is not attractive and there is one colour is used too much.

Motivation

- Management ★★★★★
- Creativity ★★★★★
- Problem solved ★★★★★
- Updates ★★★★★
- Achievment ★★★★★

" It is hard to find books related to chemistry"

User 2 – Inuri Gamage

INURI GAMAGE

Fashion Designer

About Me

Let me introduce myself, I am a fashion designer who has 3 years of experience in this job. I am very experienced in various fashion styles for all seasons and I have made various models for men and women for a famous brand. I have been involved in fashion designing since 2017 and graduated with a special degree in fashion designing from NSBM Green University.

Work Experience

2020 - 2022

FASHION DESIGNER

Picaso PVT Ltd Company, Katunayaka

Goals

- Find Fashion Designing Books.
- Enhancing knowledge of fashion design.
- Discover newly released novels and translations.
- Order required books and spend my free time with books.

Strengths

Flexibility, Patience, Dedication, Learner, Collaboration

Interest

Fashion Designing Books

Novels

Translated Books

Magazins



Female

Age - 26 Years

hello@reallygreatsite.com

123 Anywhere St., Galle,
Sri Lanka

Pain Points

- Newly released books are hard to find on this site.
- Pictures and more details are not included in the respective book item.
- There is no attraction on the site.

Motivation

- Management ★★★☆☆
- Creativity ★★★★☆
- Technology ★★★☆☆
- Updates ★★★★☆
- Budgeting ★★★★☆

"It is very difficult to find books that suit my needs."

User 3 – Kamal Bandara

KAMAL BANDARA

Student (Undergraduate at SLIIT)



About Me

Kamal is a 3rd year Information technology undergraduate student at SLIIT. He is committed to learning and reading books in free time to release his stress free. Also, he wants to improve the knowledge related his degree. He does explore so many things in internet when he wants to and always in update. As an undergraduate student he can balance time between degree works and reading other books.

male

Age - 24Years

kamalbandara@gmail.com

Katugastota rd.,
Kandy,
Sri Lanka.

Personality

- feeler
- Thinking
- scholarly

Goals

- Maintain 3.5 GPA.
- improve skill in information Technology.
- Discover newly released books.

Pain Points

- There is no books and items in same category.
- Some Educational Books are not in in there.

Strengths

Dedication,
Collaboration,
excellent verbal communication Skil

Motivation

- Management ★★★★★
- Creativity ★★★★★
- Problem solved ★★★★★
- Achievment ★★★★★

Interest

Information Technology Books
Translated Books
Magazines
English Books

Usability Issues

IT20649494 – B.D.R.A.M.A.K.B. Ranasinghe

Magazine page	Return Policy	Account (CRUD)
• Font families are same for all. • There are not proper images for related categories. • Same color is used all the items. • Not user friendly. • The facts of the content are not attractive enough to retain the customer. • Content is not attractive and there is only one color is used too much. • Popup messages are not displayed.	• Lack of clarity of content and lack of proper spacing between paragraph. • Font size is very small • same color is used everywhere. • Not user friendly • Facts of the content are not attractive enough to retain the customer.	• The form is not in good organized way. • Same font size use all over. • Not user attractive. • Popup messages are not displayed. • Could not change the user account image. • Some links are not working. • Unwanted text displayed. • Icons are not attractive and could not find easily. • Page is not well organized. • Basic design methods are allowed in account page.

IT20666842 – M.W.K.G.B. Thennakoon

Sinhala books page	About Us page	Address (CRUD)
• Customers' attractiveness in the item decreases because there are not proper images for related categories. • The content's facts are not appealing enough to keep the customer. • Font sizes are same. • When move on the item, cannot see item in proper layout. • Not user friendly.	• The information on the page is disorganized, making it difficult for users to understand this website clearly. • Because there are too many white spaces, the color contrast is unappealing. • The content's facts are not appealing enough to keep the customer. • There are no images. • It’s not user friendly • The form layout is poor.	• The form is not well-organized. • The font colors are not clear • Font size is very small. • It is not user friendly because the button background color and page background color are the same. • It is less appealing

IT20637682 – D.G.K.P. Kumara

Secondhand books page	E-Gift card page	Checkout page (CRUD)
• Alignment are not well design. • There are books without cover page images. • Navigation bar is not design properly. • Description of Some books are not included. • Font size is small and hard to find the secondhand books in navigation bar. • It is hard to understand the content of the pages.	• It's not user friendly. • Gift card is large according to the pages • Images are not included. • Interface is not well design.	• Less attractive not user friendly. • font color is not clear and different sizes. • The form is not in good organized way. • Have to fill too many steps for checkout and place order. • Card payment page is created very poorly.

IT20646592 – R.D.D. Sampath

Order page	FAQ page	Home (CRUD)
• Not user friendly. • A proper font style is not used for the description. They used low-contrast text on the Order page, so it might be hard to read or inspire the users. • Some books cover images are not displayed. • Same font use too much. • According to the "Related items" category, some cover pages of books are not displayed.	• It's not user friendly • The page is not well-organized. • Font sizes are different. (Some text too small and some text too large) • Same font size use all over. • The biggest problem is that page doesn't use a grid. • Users can not read FAQ easily and can't identify separately. (All question and answers in line by line. no white spaces) • No proper color combination.	• The page is not well-organized. • User can't identify the "Add cart" as a button. • Content is not attractive and there is only two color are used too much. • Dissonant navigational structure. • Logo style is outdated and does not attract users. • They haven't created a proper footer. • There are some of unrelated categories. • Navigation bar is Complicated to identify.

IT20648640 – W.M.S.L.K. Welgama

Discounted Books Page	Contact Us Page	Cart Page (CRUD)
• Users cannot easily find discounted books page in the navigation bar. • It is difficult for the buyers to understand about the books clearly because the information in the respective books is not well explained or organized. • Many books do not include cover images and most of the images added are not clear enough to attract buyers to purchase the books. • They have not displayed the original price of the books before marking the discount price. • The books are not classified or highlighted on the discounted books page, so the buyer cannot easily understand the book category.	• A proper font style is not used for the description. They used low-contrast text on the Contact Us page, so it might be hard to read or inspire the users. • They haven't created a proper footer. • The included contact information and links are not user-friendly. Emails cannot be accessed directly by clicking on links. • They have not used a refreshing color theme for the page. It will not attract users. • Too many white spaces.	• The cart page is not aligned properly. • Too many white spaces • The product delete icon style is outdated. • cart page is not well designed. • The background and theme are a bit boring. • No footer. • The header also looks messed up.

Scripts for contextual inquiry

Background description of the user	Questions to be asked from the user	Tasks to be given to the user
Prabhani Bhagya Prabhani Bhagya is Chemical Analyst over 2 years and she is well experienced working women. She is committed to learning and always focuses on what she is doing. She is very passionate about learning new things not only her subject but also about other visions. She is very interested to find chemical and biological books to enhance her knowledge and experience.	▪ Why do you use this site? ▪ Did you found what you are looking for after you enter the home page? ▪ Did you able to log in to website? ▪ Did you access to the account from navigation bar? ▪ Did You able to update your profile? ▪ Did you able to reset your password? ▪ Did you able to logout from the system? ▪ Are you satisfying with the login form format?	**Task 1** - Navigate to www.books.lk **Task 2** – Try to sign up for books.lk website. **Task 3** -Try to access (Login) to the created account. **Task 4** - Try to update profile details **Task 5** - Try to access orders after updating the account details. **Task 6** - Try to logout from the system.
Inuri Gamage Let me introduce herself, She is a fashion designer who has 3 years of experience in this job. She is very experienced in various fashion styles for all seasons and She has made various models for men and women for a famous brand. She approaches every project carefully, creatively and believes that her job is to blend beauty with updated styles and try to do her best within budget.	▪ How did you know about this site? ▪ Which times do you visit this kind of site? ▪ What are the product categories that you are searching mostly from this website? ▪ Have you ordered online books on this website earlier? ▪ Are you a frequent visitor to this site? ▪ Do you have any	**Task 1** - Navigate to www.books.lk **Task 2** – **Log in to your account on the website.** **Task 3** - Try to access products in the navigation bar. **Task 4** - Try to find the product type from the products list. **Task 5** - Try to add a product to the cart. **Task 6** - Try to update a product in the cart.

She has been involved in fashion design since 2017 and graduated from NSBM Green University with a special degree in Fashion Designing. She currently works at Katunayake Picasso Company.	membership on this website? ▪ Are you satisfied with this site?	**Task 7** - Try to delete a product from the cart. **Task 8** – Try to Check out the order. **Task 9** – Try to log out.
Kamal Bandara Kamal Bandara is an undergraduate at SLIIT University. He is scholarly person and very interesting to read books in free time. As a 3rd year student he wants to improve a knowledge about his degree and get same comfort to mind. So, Kamal use books.lk website to find some books, novels, magazines and order books some of items related for the degree.	▪ How often do you visit this website?? ▪ Did you find what you are looking for after you login the website? ▪ How are you find details about newly added books? ▪ Have you ordered online books on this website earlier?	**Task 1** - Navigate to www.books.lk and Create new Account. **Task 2** – Find out books in home page or Search books in search pane. **Task 3** - Try to Go to quick view in desired book. **Task 4** - Try to add desired book to cart and try to Update / Delete books.

User Interview video Links

User 1- Prabhani Bhagya

https://drive.google.com/file/d/18mzuDMpquuoofD4isXIrsWYMyq96aHs9/view?usp=sharing

User 2 – Inuri Gamage

https://drive.google.com/file/d/18BrAMFZw52s1XgaYV8CzamxpkGYW6szd/view?usp=sharing

User 3 – Kamal Bandara
https://drive.google.com/file/d/1BSFl1nP1BWeXx6XHgqmIwVY_6ZZoKnnc/view?usp=sharing

Ratings of the Usability issues identified based on contextual inquiry

Image	Problem Identified / Negative Feed back	Reason for Identified Problem	Severity (High/ Medium/ Low) and Justification for your rating	Way(s) to rectify and any Tradeoffs (i.e., Why suggested fix might not be a perfect solution)
Figure1	It will be taking more time to find the icon for unregister user to register for the system firstly.	Because of the user-unfriendly interface and not well design.	High – User confused.	We can create user friendly interface for the page.
Figure2	The whole system uses only one color and it is very difficult for the new user.	User unfriendly mismatch color themes effect.	High– Applying colors that are tiring on the eyes does not create pleasantness and it makes fed up to stay longer.	We can apply more user-friendly color themes by redesigning the interface.
Figure 3	some books do not have a front page.	Not well design the interface to attract the user.	High – Since some books do not have a front page, there is a high possibility that a person who first comes to the system will not buy such books.	Applying clear cover pages to the books that without cover pages.

Figure 4	No pop message is displayed to determine if it is registered properly.	Did not design the pop messages.	Medium – redirect to a dashboard page.	Create pop up messages after clicking login and registration buttons.
Figure 5	It is difficult to find out where the user's information is changed.	Because of the user-unfriendly interface not proper design.	Medium – User unable to understand at once where the option to update his/her account.	Redesign the interface by applying more user-friendly font groups.
Figure 6	Registration and logging forms are not clear.	Did not designed the registration and logging forms in user-friendly and not well design.	low – User unfriendly form design.	Create registration and logging forms more attractively and clear way.
Figure 7	Description of some books are not displayed.	Interface is not well organized.	High– user could not identify the related books properly.	Include the proper description of books that related to the cover page with using user-friendly method.
Figure 8	There is no pop-up massages display, when books added, update or delete from the cart.	Not develop the Pop-up massages.	Medium – difficult to find when book is added, update or delete from the cart. It makes user more confuse.	Develop the pop-up massages in correct way.
Figure 9	Can't identify the profit among the discounted price and the original price of a discounted books page.	The original price is not displayed. So, the customer could not correctly identify the profit of the book.	Medium – un user-friendly interface design.	Redesign the page with correct description with using user-friendly interface.
Figure 10	It is difficult to understand the categories on the navigation bar.	Categories are not well design in the navigation bar.	High – user is confused.	Re-design the navigation bar with using correct color format and using user friendly interface design in an attractive way.

Figure 11	Cart page is not clear. Hard to get to know the process of the page.	Cart is not design well and it is not user-friendly.	Low – time consuming.	We can re-design the cart page in more attractive way and user-friendly way.
Figure 12	Links in the contact page do not redirect to the intended page.	Interface is not well design properly.	Medium – it could not get a proper understand of the page and customer is going to confuse because of the un user-friendly interface design.	Recreate the page more user-friendly by using the Links redirected properly to the related pages and proper places.

Appendix

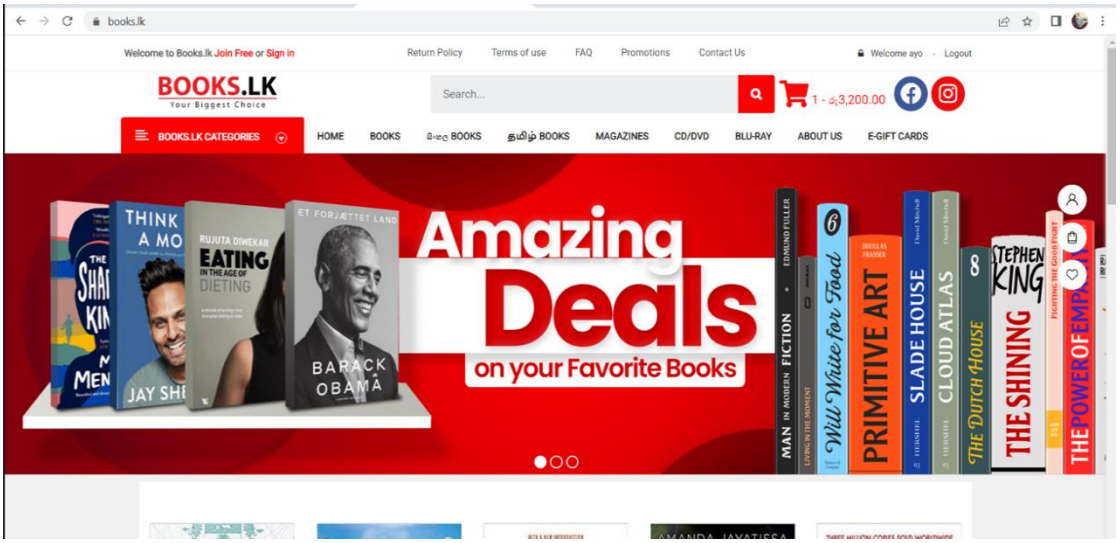
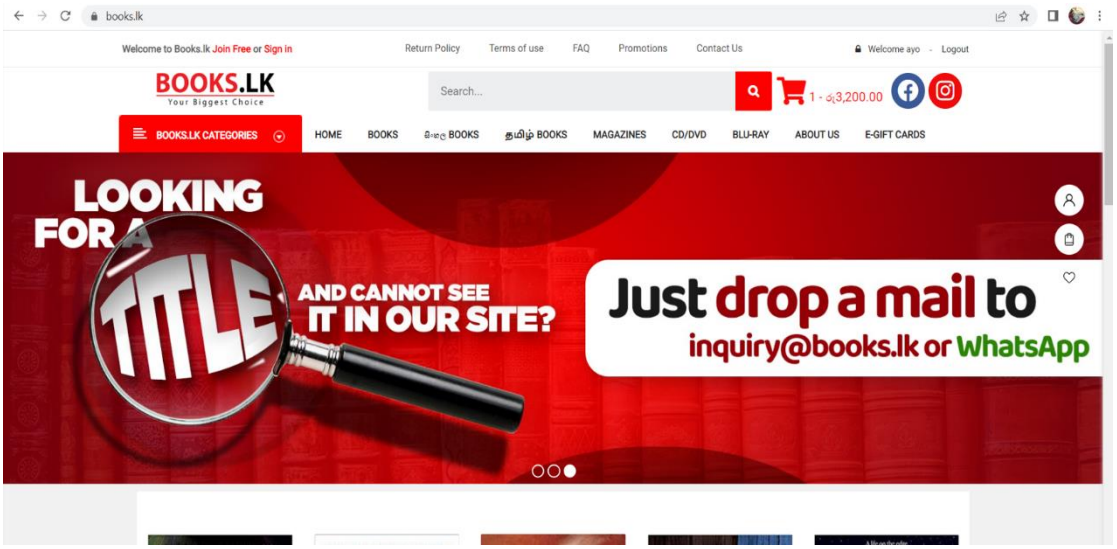
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Figure2	

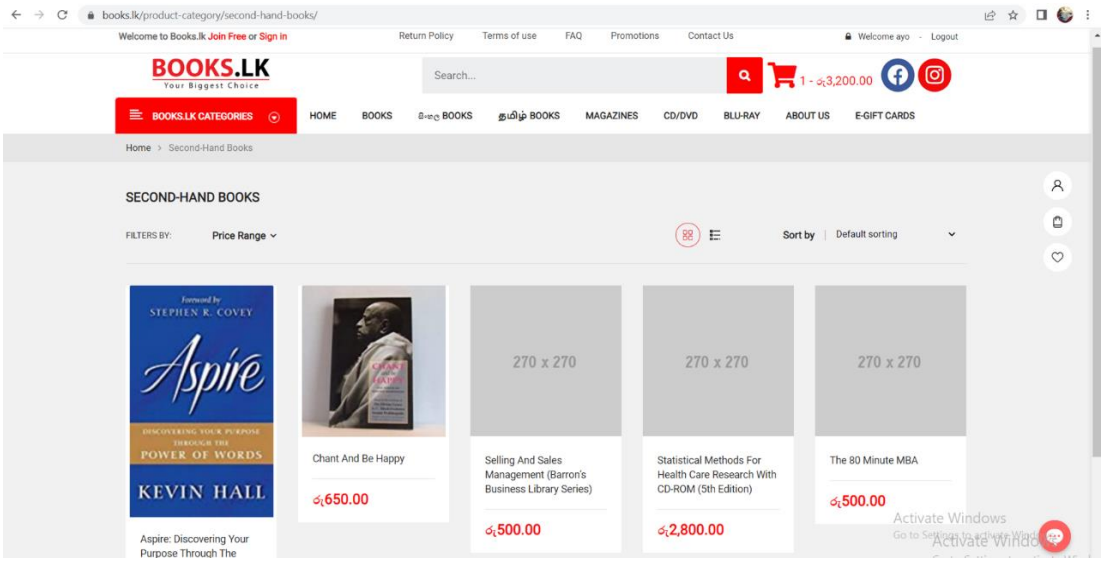
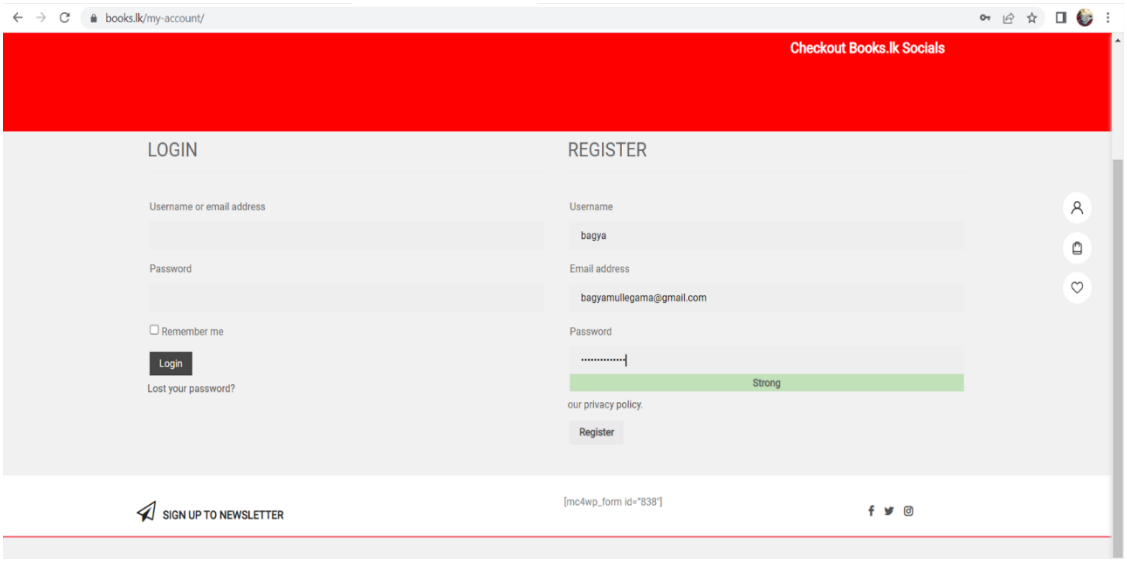
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Figure 4	

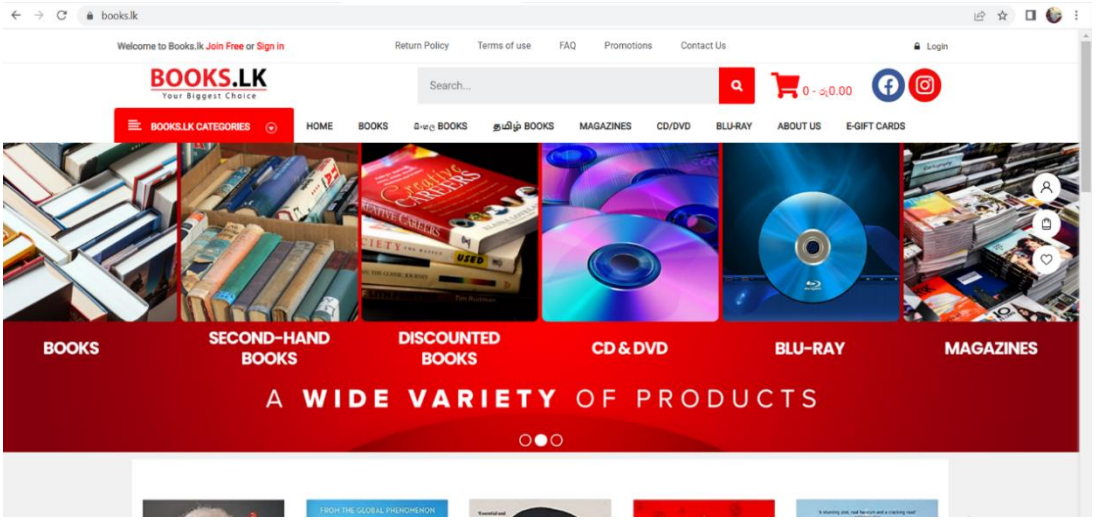
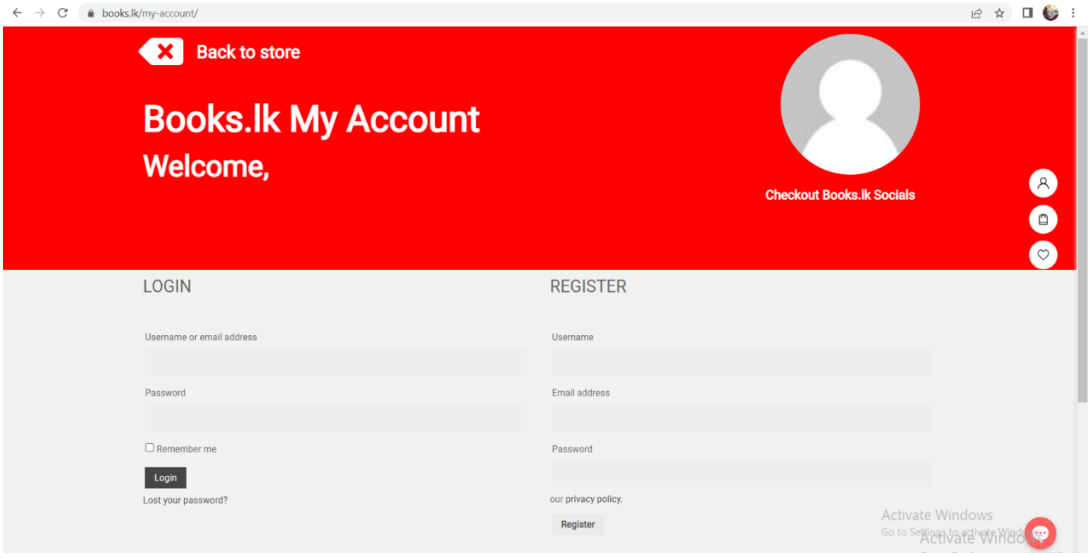
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Figure 6	

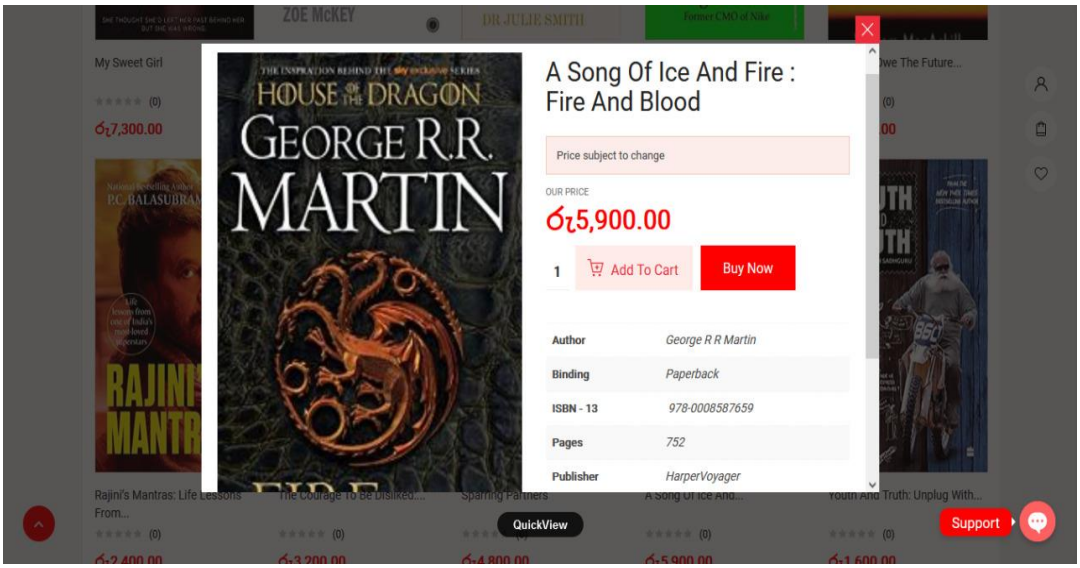
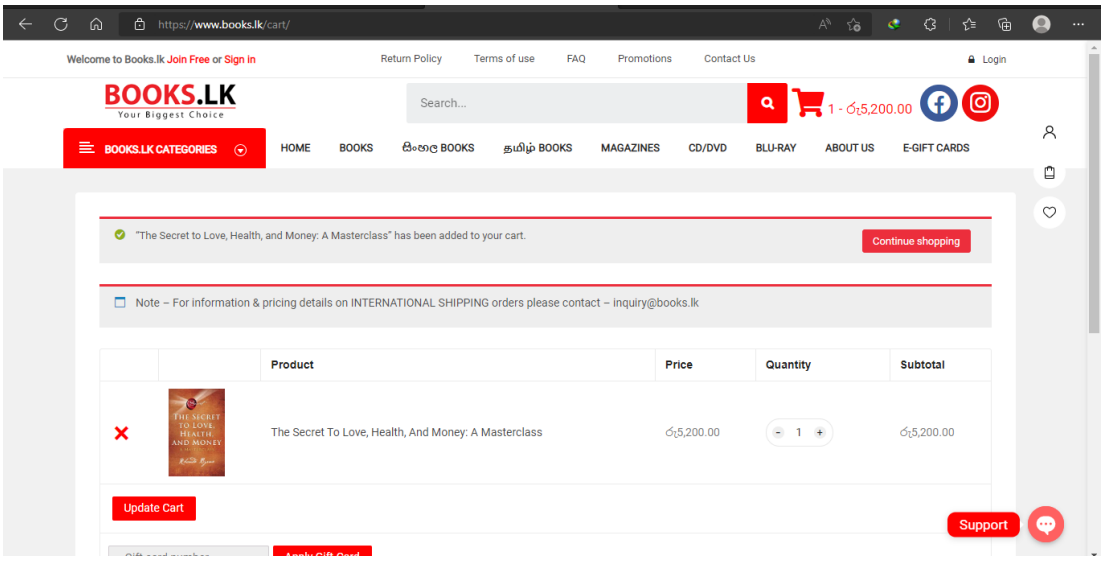
Figure Number	Figure
Figure 7	
Figure 8	

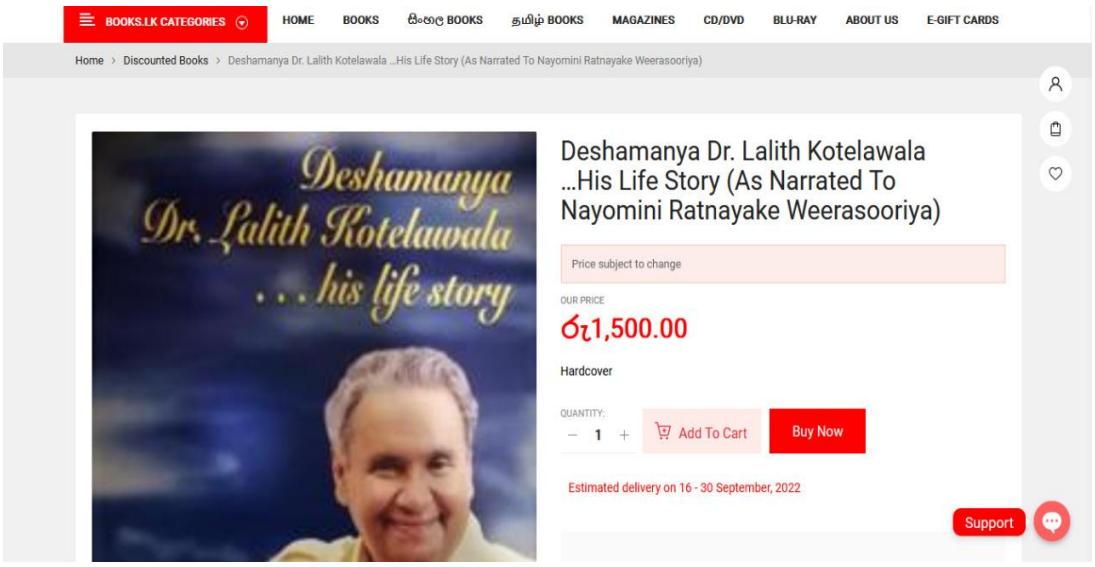
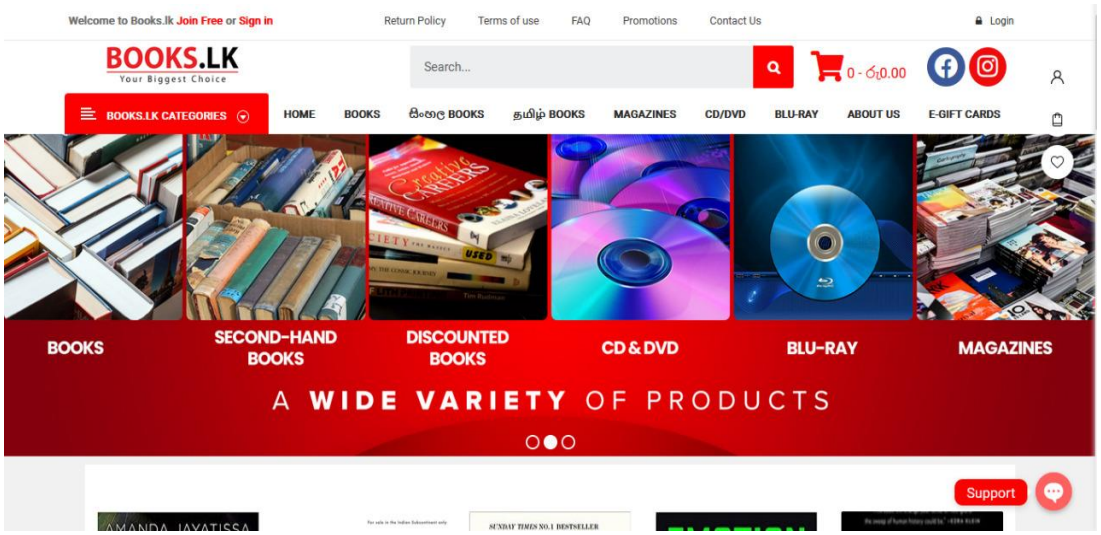
Figure Number	Figure
Figure 9	
Figure 10	

Figure 11

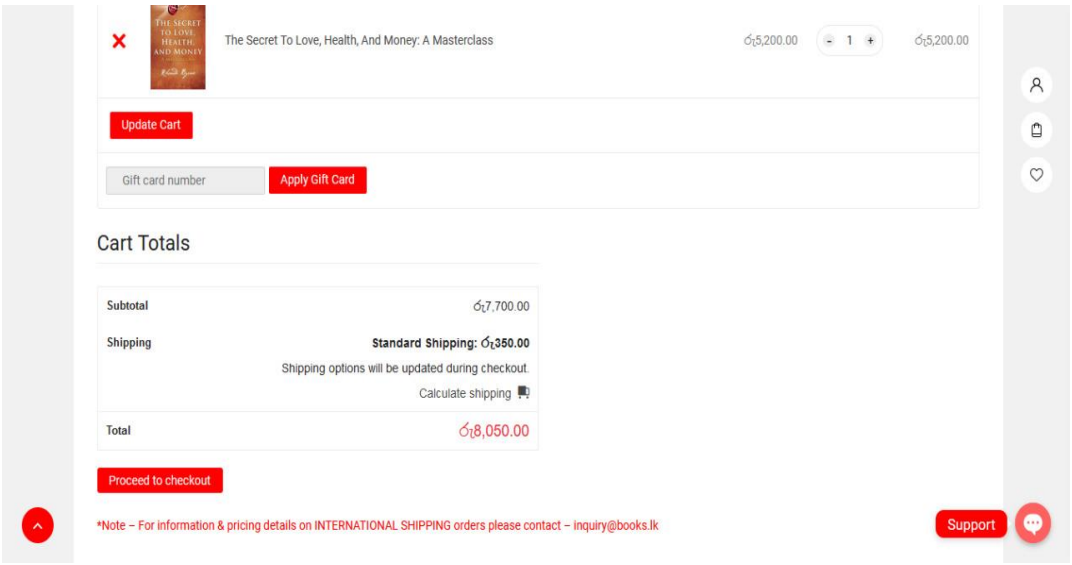
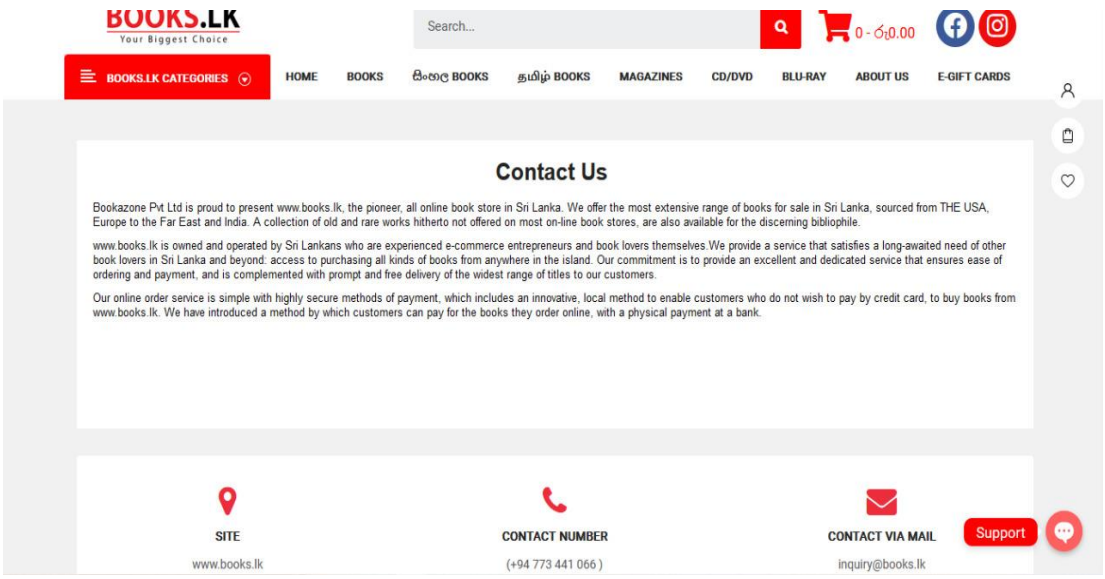
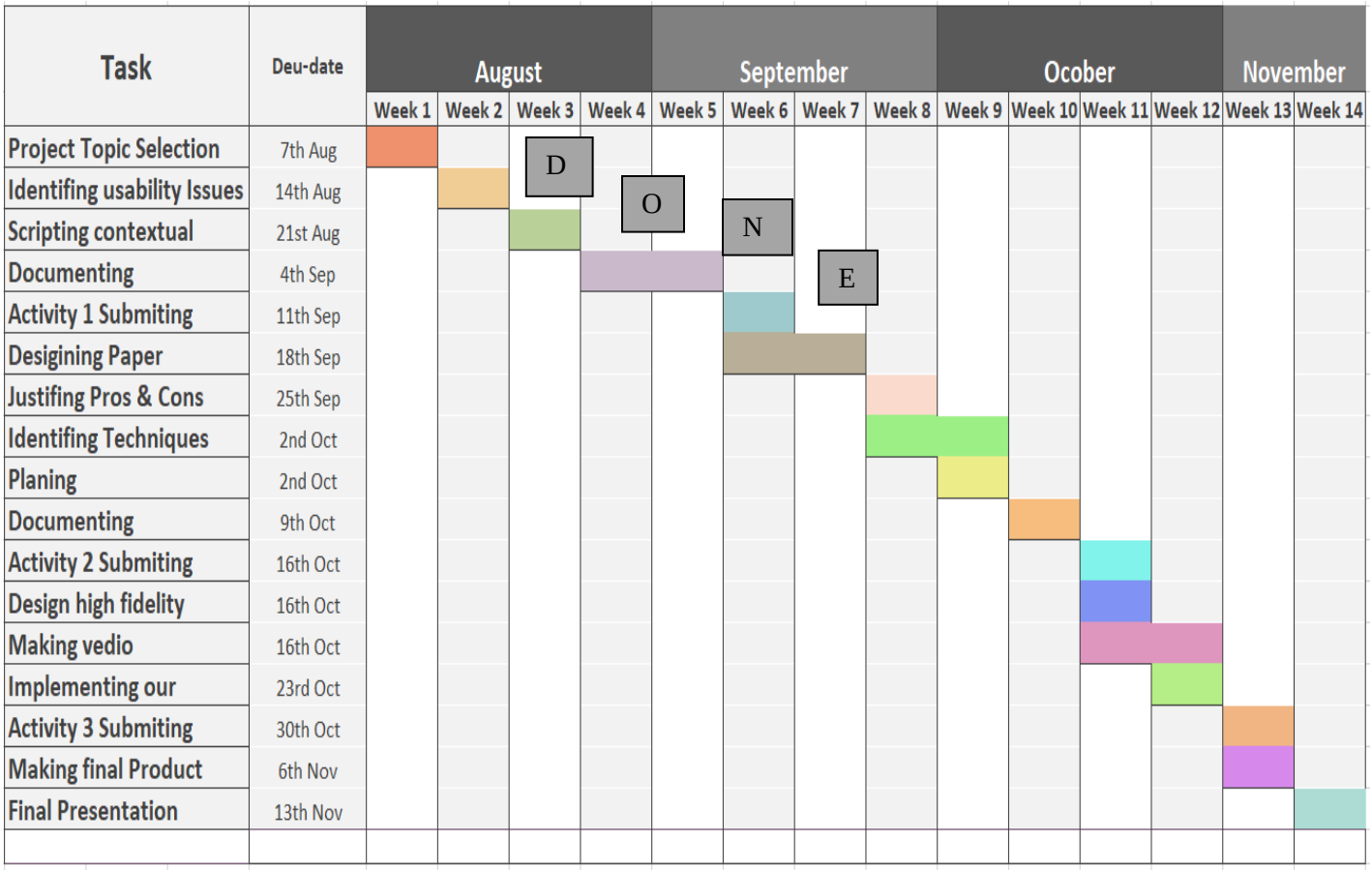


Figure 12



Time schedule (Gantt chart)



References

https://thomasdigital.com/bad-websites

https://www.allianceinteractive.com/blog/bad-websites-examples-worst-websites-designs-to-avoid/

https://www.youtube.com/watch?v=9wQkLthhHKA

https://youtu.be/Mw9LfJC9HCM

https://blog.hubspot.com/website/bad-vs-good-design

...END...

Figure10